

Deep Learning: Using Algorithms to Make Machines Think

Deep learning is part of the broader family of machine learning methods. It was introduced with the objective of moving machine learning closer to its main goal—that of artificial intelligence.



The human brain has evolved over many, many years and is one of our most important organs. The brain perceives every smell, taste, touch, sound and sight. Many decisions are taken by the brain every nano second, without our knowledge.

Having evolved over several thousands of years, the human brain has become a very sophisticated, complex and intelligent machine. What was not possible even as a dream during the 18th century and the beginning of the 19th century has become child's play now in terms of technology. Many adult brains can recognise multiple complex situations and take decisions very, very fast because of this evolution. The brain learns new things very fast now and takes decisions quickly, compared to those taken a few decades ago.

A human now has access to vast amounts of information and processes a huge amount of data, day after day, and is able to digest all of it very quickly.

Our brain is made up of approximately 100 billion nerve

cells, called neurons, which have the amazing ability to gather and transmit electrochemical signals. We can think of them as the gates and wires in a computer. Each of our experiences, senses and various normal functions trigger a lot of neuron based reactions/communications. Figure 1 shows the parts of a basic neuron.

The human brain and its neural network have been the subject of extensive research for the last several years, leading to the development of AI and machine learning technologies. The decade-long dream of building intelligent machines with brains like ours has finally materialised. Many complex problems can be now solved using deep learning techniques and algorithms. The simulation of human brain-like activities is becoming more plausible every moment.

How different is deep learning compared to machine learning

Machine learning was defined by Arthur Samuel as, "The